

ChatGPT

Valuation Metrics for IM

explain valuation Metrics Spearman Correlation Coefficient, Kendall tau, Imprecision Function, Monotonicity Index, Network connectivity in very detail with proper theory and foomulas, anology, numerical examples, importance in influence maximisation in multilayer graph.2



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Valuation Metrics for Influence Maximization in Multilayer Networks

When researchers propose a new influence-ranking algorithm (Degree, PageRank, K-shell, VoteRank, Collective Influence, etc.), they must evaluate:

"How good is the ranking produced by the algorithm?"

The actual spreading process is expensive to simulate, especially in **multilayer networks** (social + communication + transportation layers). Therefore, several evaluation metrics are used to compare rankings.

The most common metrics are:

1. Spearman Correlation Coefficient (ρ)
2. Kendall Tau (τ)
3. Imprecision Function
4. Monotonicity Index
5. Network Connectivity

1. Spearman Correlation Coefficient (ρ)

What does it measure?

Measures the similarity between two rankings.

Suppose:

- Ranking A = produced by your algorithm
- Ranking B = true ranking obtained from influence spread simulations

Spearman tells us:

"How closely does our ranking follow the true ranking?"

Analogy

Imagine a class of students.

Actual Rank

Student	Actual Rank
A	1
B	2
C	3
D	4
E	5

Predicted Rank

Student	Predicted
A	1
B	3
C	2
D	4
E	5

Almost identical.

Spearman will be close to +1.

Formula

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where:

- d_i = difference between ranks
- n = number of nodes

Numerical Example

Actual Ranking

Node	Rank
A	1
B	2
C	3

Node	Rank
D	4
E	5

Predicted

Node	Rank
A	1
B	3
C	2
D	4
E	5

Compute:

$$d_A = 0$$

$$d_B = 2 - 3 = -1$$

$$d_C = 3 - 2 = 1$$

$$d_D = 0$$

$$d_E = 0$$

$$\sum d_i^2 = 0 + 1 + 1 + 0 + 0 = 2$$

$$\rho = 1 - \frac{6(2)}{5(25 - 1)}$$

$$\rho = 1 - \frac{12}{120}$$

$$\rho = 0.9$$

Interpretation

Value	Meaning
+1	Perfect agreement
0	No correlation
-1	Opposite ranking

Importance in Influence Maximization

Suppose:

- True influential nodes from SIR simulation
- Estimated influential nodes from centrality measure

High Spearman means:

✓ Centrality measure accurately predicts influence

Low Spearman means:

✗ Poor ranking quality

2. Kendall Tau (τ)

What does it measure?

Measures pairwise ranking agreement.

Instead of comparing rank differences, it asks:

"For every pair of nodes, did we place them in the correct order?"

Analogy

Suppose:

Actual:

$$A > B > C > D$$

Predicted:

$$A > C > B > D$$

Only B and C swapped.

Most pairs remain correct.

Thus Kendall Tau remains high.

Concordant and Discordant Pairs

Concordant

Both rankings agree.

Example:

Actual:

$$A > B$$

Predicted:

$$A > B$$

Discordant

Order reversed.

Actual:

$$A > B$$

Predicted:

$$B > A$$

Formula

$$\tau = \frac{C - D}{\frac{n(n-1)}{2}}$$

Where

- C = concordant pairs
- D = discordant pairs

Numerical Example

Ranking:

$$A > B > C$$

Predicted:

$$A > C > B$$

Pairs:

(A,B) ✓

(A,C) ✓

(B,C) ✗

Therefore:

$$C = 2$$

$$D = 1$$

Total pairs:

$$3$$

$$\tau = \frac{2 - 1}{3}$$

$$\tau = 0.333$$

Difference from Spearman

Spearman	Kendall
Rank distance	Pairwise order
More sensitive to large rank shifts	More robust
Faster	More reliable

Importance in Influence Maximization

Two algorithms may have similar Spearman values.

Kendall Tau reveals:

Which one preserves the ordering of influential nodes better.

Especially important when selecting top-k seed nodes.

3. Imprecision Function

This is one of the most important metrics in influence maximization.

Purpose

Measures:

"How much influence is lost when we choose seeds using a centrality measure instead of the true best spreaders?"

Idea

Suppose:

True Top-5 Spreaders

$$\{1, 4, 7, 10, 15\}$$

Spread:

$$M_{eff} = 500$$

Algorithm predicts:

$$\{1, 3, 7, 9, 15\}$$

Spread:

$$M_{alg} = 450$$

Loss:

$$500 - 450 = 50$$

Formula

$$\varepsilon(p) = 1 - \frac{M_{alg}(p)}{M_{eff}(p)}$$

Where:

- p = fraction of top nodes
 - M_{eff} = influence of actual best spreaders
 - M_{alg} = influence from ranking algorithm
-

Example

True spread:

$$M_{eff} = 1000$$

Algorithm spread:

$$M_{alg} = 900$$

Then:

$$\varepsilon = 1 - \frac{900}{1000}$$

$$\varepsilon = 0.1$$

10% influence loss.

Interpretation

Value	Meaning
0	Perfect
Small	Good
Large	Poor

Importance in Multilayer Networks

In multilayer networks:

- Influence can travel through many layers.
- Wrong seed selection causes huge spread reduction.

Thus Imprecision directly measures practical effectiveness.

Researchers often plot:

Imprecision vs Top-k %

Lower curve = better algorithm.

4. Monotonicity Index

Problem

Many centrality measures produce ties.

Example:

Node	Score
A	10
B	10
C	10
D	8

Now:

Who is first?

Who is second?

Ambiguous.

Purpose

Measures ranking resolution.

Can the method uniquely distinguish nodes?

Analogy

Two exams:

Exam A

Marks:

95,94,93,92,...

Everyone different.

Easy ranking.

Exam B

Marks:

90,90,90,90,90

Impossible to rank.

Exam A has higher monotonicity.

Formula

If

$$N$$

nodes.

Group sizes:

$$n_1, n_2, \dots, n_m$$

Then

$$M = \left(1 - \frac{\sum n_i(n_i - 1)}{N(N - 1)} \right)^2$$

Example

Suppose

10 nodes

Groups:

4, 3, 2, 1

Then

$$\begin{aligned}\sum n_i(n_i - 1) &= 4(3) + 3(2) + 2(1) + 1(0) \\ &= 20 \\ M &= \left(1 - \frac{20}{90}\right)^2 \\ &= 0.605\end{aligned}$$

Interpretation

Value	Meaning
1	Unique ranking
0	All tied

Importance

Influence maximization requires selecting:

Top 1%

Top 5%

Top 10%

If many nodes have identical scores:

- Seed selection becomes unstable.
- Different tie-breaking gives different spreads.

Higher monotonicity means better discrimination.

5. Network Connectivity

This metric evaluates the structural integrity of the network.

Definition

Connectivity indicates:

How well nodes remain connected with each other.

In multilayer networks, connectivity determines how easily influence can propagate.

Analogy

Imagine cities connected by roads.

Highly Connected

Many alternate routes.

If one road fails:

Traffic still moves.

Poorly Connected

Single bridge.

Bridge fails.

Everything stops.

Influence behaves similarly.

Common Connectivity Measures

(A) Largest Connected Component (LCC)

Largest group of mutually reachable nodes.

Formula:

$$LCC = \frac{\text{Nodes in largest component}}{\text{Total nodes}}$$

Example

Network:

100 nodes

Largest component:

80 nodes

$$LCC = \frac{80}{100} = 0.8$$

(B) Average Degree

$$\langle k \rangle = \frac{2E}{N}$$

where:

- E = edges
 - N = nodes
-

Example:

$$N = 10$$

$$E = 20$$

$$\langle k \rangle = 4$$

(C) Algebraic Connectivity

Second-smallest eigenvalue of graph Laplacian.

$$\lambda_2$$

Higher:

$$\lambda_2 \Rightarrow \text{stronger connectivity}$$

Importance in Influence Maximization

Influence spreads only through connections.

If connectivity is weak:

- Even perfect seed selection fails.

If connectivity is high:

- Cascades become easier.

Researchers often examine:

- Connectivity before attack
- Connectivity after removing influential nodes

A good influential-node ranking often causes a rapid drop in connectivity when those nodes are removed.

Why These Metrics Matter Together

Metric	Measures
Spearman	Similarity of rankings
Kendall Tau	Pairwise ordering accuracy
Imprecision Function	Actual spreading loss
Monotonicity Index	Ranking resolution
Network Connectivity	Structural ability to spread influence

In Multilayer Influence Maximization

Consider a multilayer network:

- Layer 1: Facebook
- Layer 2: WhatsApp
- Layer 3: Twitter
- Layer 4: Physical contacts

A proposed centrality method identifies influential users.

Evaluation workflow:

1. Generate ranking using proposed method.
2. Run diffusion model (IC/SIR/LT) to obtain true influence.
3. Compute Spearman (ranking similarity).
4. Compute Kendall Tau (pairwise consistency).
5. Compute Imprecision Function (spread loss).
6. Compute Monotonicity (ability to distinguish users).
7. Analyze Connectivity (how influence pathways exist across layers).

A strong influence-maximization method typically shows:

- **High Spearman (≈ 1)**
- **High Kendall Tau (≈ 1)**
- **Low Imprecision (≈ 0)**

- **High Monotonicity (≈ 1)**
- **Strong multilayer connectivity utilization**

These metrics together provide a complete evaluation of whether a centrality measure truly identifies the most influential spreaders in complex multilayer networks.